

UCLA Undergraduate
RESEARCH WEEK

UCLA College | Physical Sciences
Statistics & Data Science



VA



U.S. Department
of Veterans Affairs

Investigating the Connection Between Self-Identified Trauma and Brain Inflammation

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Presentation Overview

- Introduction
- Research Question
- Context
- Investigation / Methods
- Findings
- Conclusion

Introduction

- Data comes from a clinical survey of $N = 125$ patients
 - Participants given the Traumatic Experience Checklist, or “TEC” (Nijenhuis et al., 2002)
 - self-report questionnaire inquiring about 29 types of potential trauma
 - Brain inflammation levels (denoted as brain total volume, V_T) were measured using Positron Emission Topography scans of the radioligand [^{18}F]-FEPPA (Mizrahi et al., 2012)
 - five brain regions: medial prefrontal cortex, dorsolateral prefrontal cortex, anterior cingulate cortex, hippocampus, and cerebellum
 - Tracked status of relevant TSPO genotype
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Research Question

- Do higher scores on the TEC have a relationship with elevated brain inflammation?
 - **No**, we could not find any relationship.

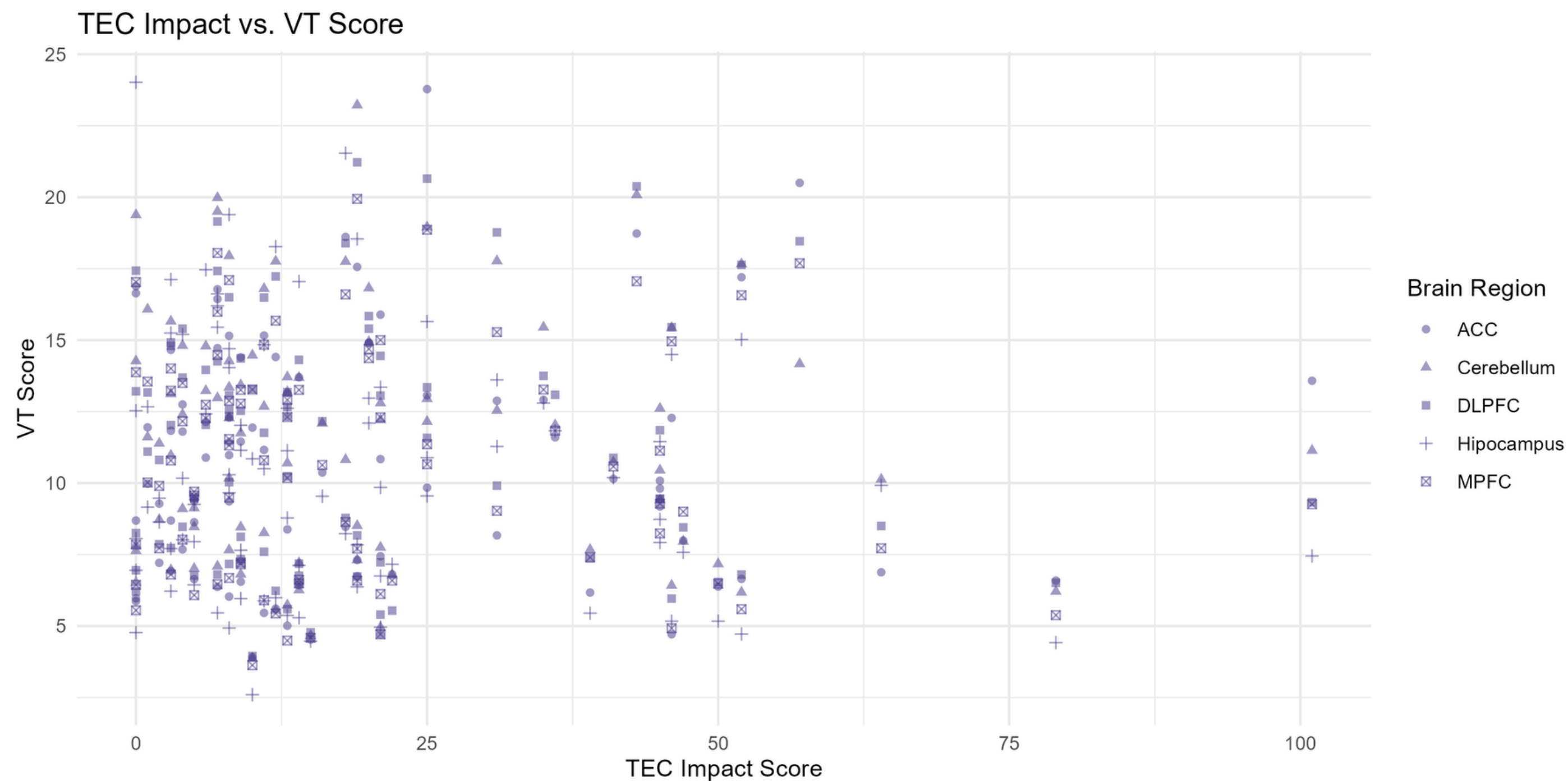
Context

- PTSD severity shown to have association with elevated brain inflammation markers (Y. Deri et al., 2021)
- Trauma and adversity during childhood exacerbates inflammation and affects brain development (A. Danese & J. Baldwin, 2017)
- Caution around assuming causality (direction) vs. correlation (Sumner et al., 2020)

Context

- Some notes on the Traumatic Experience Checklist:
 - “... the TEC is not a diagnostic tool,” but rather “a self-report measure addressing **potentially** traumatising events.”
 - Not an objective measure of trauma, but rather an individual’s subjective experience of their environment
 - (<https://www.enijenhuis.nl/tec>)
- Indicator `TEC\_impact\_total`

Investigation

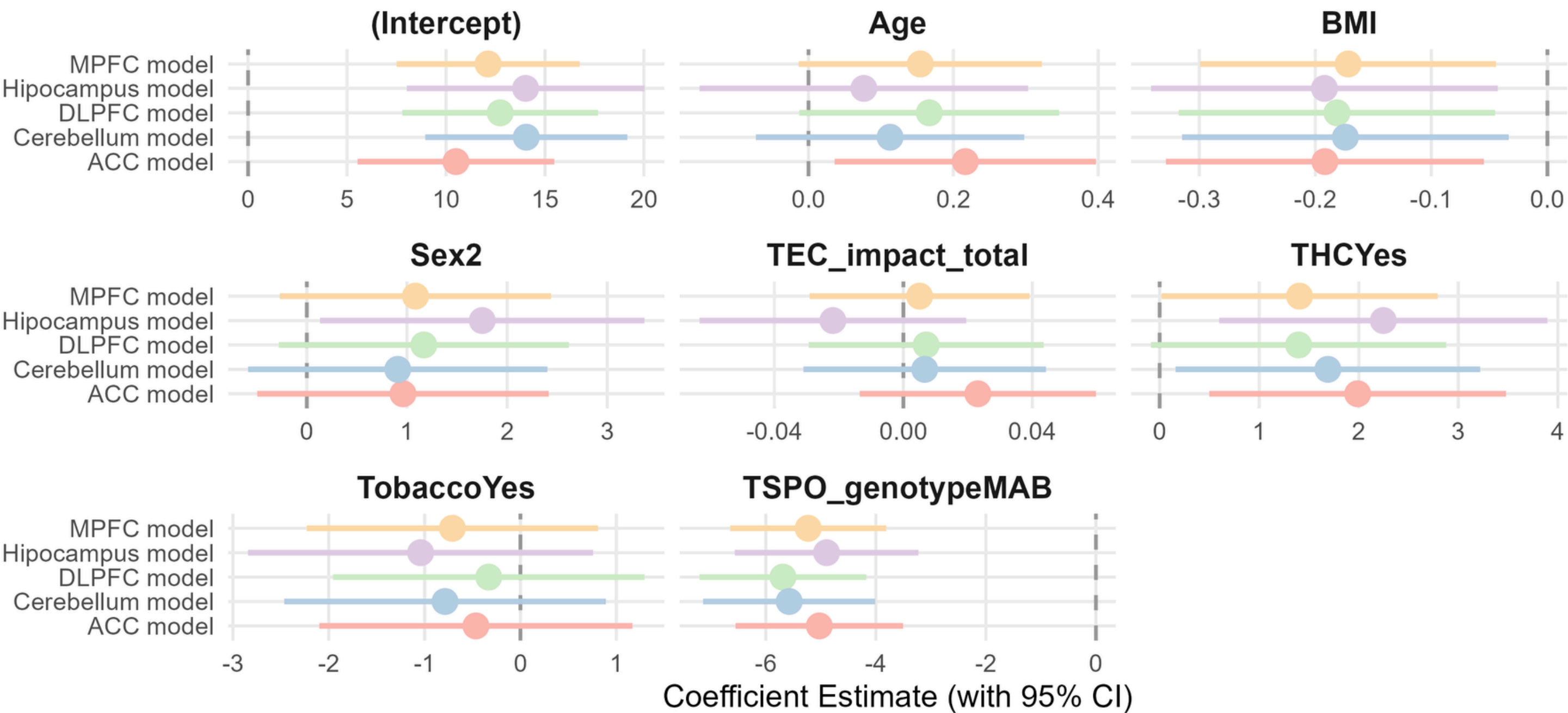


Investigation

$$\begin{aligned} VT_score = & \beta_0 + \beta_1(TEC_impact_total) + \beta_2(Age) + \beta_3(BMI) \\ & + \beta_4(Sex) + \beta_5(THC) + \beta_6(Tobacco) + \beta_7(TSPO_genotype) + \epsilon \end{aligned}$$

Findings

Parameter Estimates Across Brain Regions



Conclusion

- did **not** find any hypothesized effect
 - strong null finding: not a large effect with insufficient sample size for significance, but rather a small effect size
 - therefore, no indication for clinical relevance
- limitations
 - self-reported (subjectivity)

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